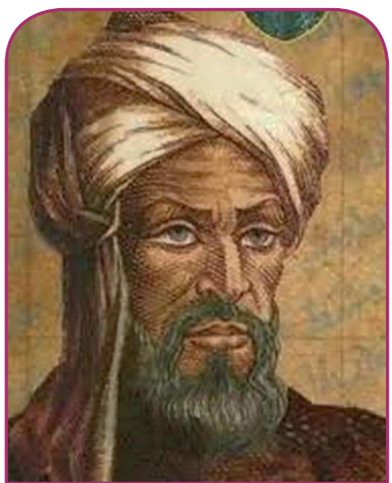
Chapter 2

ALGEBRA

Learning Objectives

After studying this chapter, the students will be able to understand

- definition of partial fractions
- the various techniques to resolve into partial fractions
- different cases of permutation such as word formation
- different cases of combination such as word formation
- difference between permutation and combination
- the concept and principle of mathematical induction
- Binomial expansion techniques



Al-Khwarizmi

Introduction

Algebra is a major component of mathematics that is used to unify mathematics concepts. Algebra is built on experiences with numbers and operations along with geometry and data analysis. The word "algebra" is derived from the Arabic word "al-Jabr". The Arabic mathematician Al-Khwarizmi has traditionally been known as the "Father of algebra". Algebra is used to find the perimeter and area of any plane region, volume and CSA of solid.

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2.1 Partial Fractions

Rational Expression

An expression of the form $\frac{p(x)}{q(x)} = \frac{a_0x^n + a_1x^{n-1} + \cdots + a_n}{b_0x^m + b_1x^{m-1} + \cdots + b_m}, [q(x) \neq 0]$ is called a rational algebraic expression.

If the degree of the numerator $p(x)$ is less than that of the denominator $q(x)$ then $\frac{p(x)}{q(x)}$ is called a proper rational expression. If the degree of $p(x)$ is greater than or equal to that of $q(x)$ then $\frac{p(x)}{q(x)}$ is called an improper rational expression.

An improper rational expression can be expressed as the sum of an integral function and a proper rational expression.

That is, $\frac{p(x)}{q(x)} = f(x) + \frac{r(x)}{q(x)}$

For example,

$\frac{x^2 + x + 1}{x^2 + 2x + 1} = 1 - \frac{x}{x^2 + 2x + 1}$

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